Seeing Is Not Believing: Counterfactual Lesion Erasure Teaches a Small Medical Vision–Language Agent to Report Only What the Image Shows

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Draft for expert review. Every number comes from the pre-registered v4 runs (notebooks 23–25) and is regenerated from stored per-view transcripts. Items in orange brackets are open: the reader study (Section 7), a few values to be pulled from the stored runs, and a final check of every reference. Not for reader-study participants until they have submitted their answers.

Abstract

Background. Medical vision–language models (VLMs) often report a finding because the question names it, not because the image shows it. Standard benchmarks rarely detect this, because the asked finding is usually present.

Methods. From public segmentation datasets we built counterfactual views without human annotation: the lesion erased by inpainting; a control in which a region of the same shape elsewhere is erased with the same tool; a degraded copy; and unedited lesion-free images. The *evidence sensitivity gap* (ESG) is the rate of "absent" answers after erasure minus the rate after the control edit; it is zero for any model that ignores the image or only detects edits. MedScope-4B, Qwen3.5-4B with LoRA and zoom and windowing tools, was trained in three stages on label-derived trajectories. Data, recipe and model-selection rule were pre-registered; the selected model was tested once against nine frontier and six open VLMs in an identical agent loop, with paired bootstrap tests.

Results. On 507 internal test cases across five imaging types, MedScope-4B reached ESG 0.758 and claimed an erased lesion in 9.6% of cases, against ESG 0.27–0.47 and 41–62% for the frontier models (all $p < 0.001$). Accuracy on real lesions did not differ from any frontier model, and false findings on unedited healthy images (7.6%) were lower than for eight of nine. Removing only the counterfactual views lowered ESG from 0.741 ± 0.019 to 0.279 ± 0.107 over three seeds and raised erased-lesion claims from 13% to 68%; the effect replicated on MedGemma 1.5 4B and on two inpainting tools never used in training. At four external centres the ESG advantage held (0.597 vs 0.28–0.41), but frontier models detected more real lesions and net utility did not differ. On a fresh external test MedScope-4B missed every pneumothorax; diagnosis traced this to a training coverage gap amplified by the final training stage.

Conclusions. Counterfactual training gives a small model an evidence-gated reading disposition that prompting does not give frontier models. It does not improve perception, and rare critical findings need explicit coverage in training and selection.

Keywords: medical vision–language models; hallucination; counterfactual data; evidence sensitivity; inpainting; pre-registration; agentic AI

Highlights

- Lesion erasure with a same-shape, same-tool edit elsewhere tests whether an answer follows the image.
- The evidence sensitivity gap is zero for every image-blind strategy, including edit detection.
- A 4B agent trained only from dataset labels claims erased lesions in 9.6% of cases; nine frontier models, 41–62%.
- Without the counterfactual views, erased-lesion claims rise to 68% (three seeds); the effect replicates on a second base model.
- A pre-registered fresh test exposed a complete pneumothorax miss, traced to training coverage.

1 Introduction

A radiologist asked "Is there a polyp?" looks for one. A vision–language model asked the same question may answer "Yes, a sessile polyp is visible" whether or not the image contains one. In the image–text data these models learn from, a question that names a finding is itself strong evidence that the finding is present, and a model can learn to answer from the question rather than from the pixels. For a clinical assistant this is the most dangerous kind of error: a confident, plausible report of something that is not there.

Medical VLMs are now built by instruction-tuning general models on biomedical data [1, 2, 3, 4, 5] and are evaluated mainly on VQA-RAD, SLAKE and PathVQA [6, 7, 8]. In these benchmarks the presupposition of a closed question is usually true, so a model that leans on the question scores well. Hallucination benchmarks [9, 10, 11, 12, 13] probe absent findings, but they change the question or the answer while the image stays the same, so they cannot tell whether a correct "No" came from the image or from a learned prior about the question.

Testing whether an answer depends on the image needs image pairs that differ only in the finding. Such pairs cannot be acquired clinically for most findings, and expert checking of synthetic pairs at scale is costly. We build them automatically from public segmentation datasets by erasing the expert-masked lesion with learned inpainting [14]. The obvious confound is that a model could learn to detect *that an image was edited* rather than *whether the lesion is there*. Every lesion-erased view therefore has a twin in which a region of exactly the same shape is erased elsewhere with the same tool, leaving the lesion intact. Only a model that reads the lesion can treat the two differently.

We use these views twice: to measure evidence sensitivity, and to train it. A 4-billion-parameter agent that can zoom and window the image is trained on trajectories derived from dataset labels alone, selected by a rule committed before any of its data existed, and tested once against nine frontier and six open models through an identical agent loop.

Contributions.

1. **Counterfactual evidence views with an edit-matched control**, built and quality-checked automatically across five imaging types, with no human annotation (Section 3.2).
2. **The evidence sensitivity gap (ESG)**, zero for every image-blind strategy (always yes, always no, always abstain, react to edits), reported with false findings on erased lesions and on unedited healthy images, grounding, and a clinical cost-matrix utility (Section 3.7).
3. **MedScope-4B**, selected by a pre-registered rule, which claims erased findings four to six times less often than nine frontier models while matching their accuracy on real lesions, robustly to prompting, question wording and unseen inpainting tools (Sections 5.1–5.5).
4. **Causal evidence on what produces the behaviour**: a three-seed ablation of the counterfactual views, an arm matched for the share of "absent" answers, an erasures-only arm and a second base model (Section 5.3).
5. **A transparent account of failure**: at new centres frontier models detect more real lesions, and a confirmatory fresh test revealed a complete miss of pneumothorax, whose cause we trace to training coverage (Sections 5.6, 6).

2 Related work

Medical vision–language models. General medical assistants are built by instruction-tuning general VLMs on biomedical image–text data: LLaVA-Med [1], HuatuoGPT-Vision [2], MedGemma [3, 4] and Lingshu [5]. They are evaluated mostly on VQA-RAD [6], SLAKE [7] and PathVQA [8], where the asked finding is usually present and accuracy rewards agreement with the question's presupposition.

Hallucination benchmarks and detectors. POPE [9] probes object hallucination with questions about absent objects. Medical benchmarks — CARES [10], Med-HallMark [11], MedVH [12] and MedHEval [13] — curate or perturb questions and answers. Two recent training-free detectors use counterfactual probes at inference: counter-evidence verification [15] blacks out the region supporting a claim and checks whether the claim persists, and counterfactual visual grounding uncertainty [16] contrasts grounding on factual and perturbed inputs. Our work differs in three ways: we change the *image* while holding the question fixed; we control the edit with a matched tool and shape, so that edit detection earns nothing; and we use the views to *train* a model, not only to audit one.

Counterfactual image editing. Agarwal et al. [17] edited natural images for VQA, removing objects so that the answer should change (covariant edits) or irrelevant objects so that it should not (invariant edits), and showed that VQA models rely on spurious correlations. Our target and random removals are the medical analogue, with two additions: the invariant edit uses the same tool and the *same shape* as the covariant one, and abstention and clarification are scored by a clinical cost matrix. Learned inpainting such as LaMa [14] makes plausible fills of large masks possible; classical inpainting [18], Poisson blending [19] and diffusion inpainting [20] provide independent tools for testing whether a model learned one tool's traces.

Tool-using and reinforcement-trained agents. Recent agents zoom or crop to "think with images" and are trained with reinforcement learning [21, 22]; in medicine, MMedAgent [23] learns to call tools and MMedAgent-RL [24] optimises multi-agent collaboration. Our agent uses two simple tools. On-policy RL (GRPO [25] with DAPO-style dynamic sampling [26]) gave no signal in pilot runs, so we use rejection-sampling fine-tuning [27, 28]. Selective prediction is summarised by the area under the risk–coverage curve [29]. We follow the CLAIM 2024 checklist [30] (Appendix C).

3 Methods

3.1 Task and agent protocol

An episode presents one image and one question whose answer depends on a lesion. Each turn the model writes brief reasoning followed by exactly one tagged JSON action (prompt in Appendix A): `zoom` (a box on the original image, integer coordinates 0–1000, returning the cropped region) or `window` (centre and width on a 0–255 scale, returning the contrast-adjusted image), at most three calls per episode; `answer`, with an optional evidence box and a stated confidence; `refer` (usable image, cannot decide); or `clarify` (unusable image or wrong anatomy). When the asked finding is not visible the model must say so ("No" to "Is there…?", "Yes" to "Is …healthy?", "No abnormality" to an open question). Every model in this paper receives the same prompt, tools and turn structure.

3.2 Counterfactual views

For every case (image, expert lesion mask, question whose answer depends on the lesion) we build four views; unedited lesion-free images from the same sources form a fifth kind (Figure 1; real test views in Figure 2).

Original. Unchanged. Expected: the correct answer with a box on the lesion.

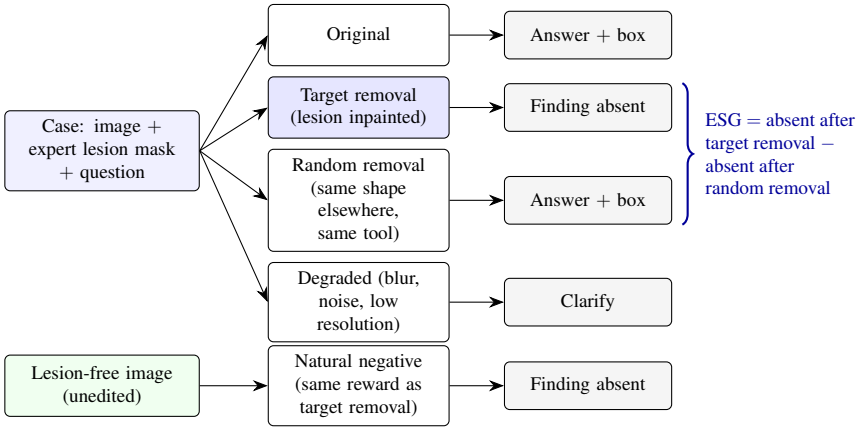


Figure 1: Views built from each case and the expected action. Target removal and the random-removal control share the editing tool and the edited shape, so a model that only notices edits treats them alike; ESG measures the difference. Natural negatives share target removal's expected answer and reward, so detecting edits earns nothing.

Target removal. The lesion mask, dilated by 5 pixels, is inpainted. Expected: the answer stating the finding is absent.
Random removal (control). The same dilated shape, moved to a random foreground position that does not touch the dilated lesion, inpainted with the same tool. No lesion pixel changes. Expected: the original answer.
Degraded. Gaussian blur ($\sigma = \max(h,w)/40$), Gaussian noise ($\sigma = 60$), $8\times$ down- and up-sampling, JPEG compression, motion blur, or an image of a different imaging type. Expected: `clarify`.
Natural negative. An unedited lesion-free image with the dataset's label. Expected: the absent answer, with exactly the reward of target removal.

Two symmetries make the design work. Target removal and natural negatives share expected answer and reward, so learning to detect edits earns nothing. Target and random removal share tool and shape, so a model that reacts to edits treats them alike and scores ESG 0. Lesions under 20 pixels or over 20% of the image are excluded. Training additionally uses lesion-insertion views (a real lesion moved into a healthy training image; expected: present) and a second erasure tool, tissue transplant (tissue from elsewhere in the same image, Poisson-blended [19]), for half of the cases.

3.3 Automated quality control

No human checks any view. A case is kept only if (i) the lesion no longer stands out after target removal [exact criterion from the code], (ii) no lesion pixel changed in random removal, and (iii) no view is a near-copy of a training image (perceptual hash, confirmed by pixel correlation). Splits are made over patients, or over groups of near-identical images where patient identifiers are missing; SLAKE's official split leaks neighbouring slices (4 of 63 validation and test images had a near-copy in training), which group splitting removes. A texture classifier trained to separate target from random edits drives adversarial filtering until its AUC is at most 0.6 (test set: 0.56). A hand-crafted shortcut classifier on the v4 training views reached AUC 0.59, so no further filtering was needed.

3.4 Reward and net utility

Each decision is scored with a clinical cost matrix (Table 1). It is the selection reward during training and the *net utility* metric at test. A correct answer on an original or random-removal view earns 0.5 more when its box overlaps the lesion with IoU ≥ 0.3 (boxes covering over half the image earn no bonus); unparseable output scores -1; each tool call beyond the first costs 0.05. Section 5.5 reports net utility under four alternative matrices.

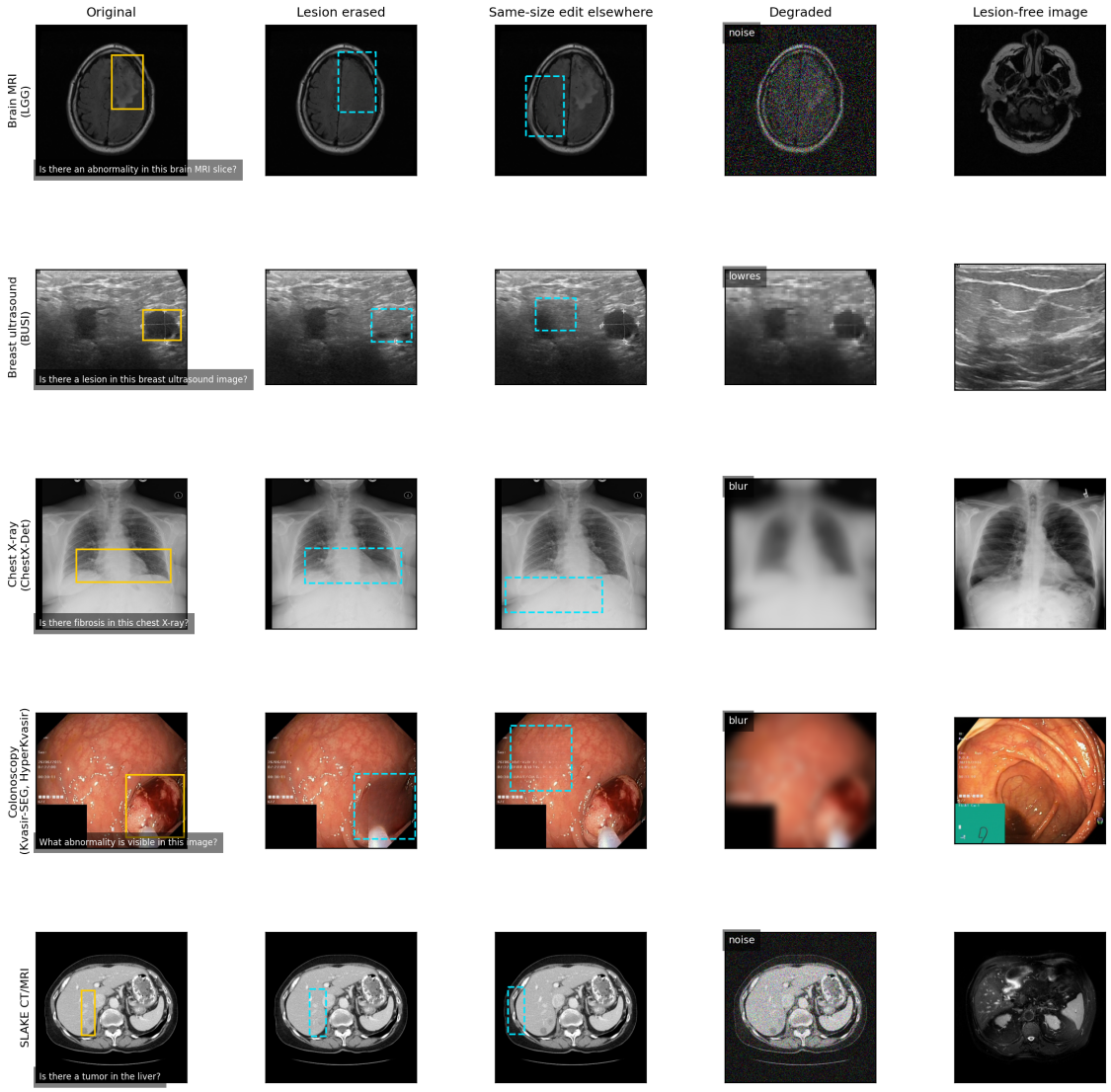


Figure 2: Real test views, one case per imaging type. Yellow: lesion (reference-standard mask); dashed cyan: edited region. Note the second dark region in the breast-ultrasound image that the mask does not cover (Section 8.5).

Table 1: Cost matrix (reward per view). Target removal and natural negatives are scored identically by design.

View	Correct / absent answer	Wrong, or claims finding	Refer	Clarify
Original, random removal	+1.0	−2.0 (−4.0 critical)	−0.3	−0.3
Target removal, natural negative	+1.0	−2.0	+0.5	−0.5
Degraded	+0.25	−2.0	0.0	+0.5

3.5 Model and training

The base model is Qwen3.5-4B [31] with LoRA [32] (rank 16, $\alpha = 32$, dropout 0.05, on all linear layers of the language model); the vision encoder is frozen throughout. Training trajectories are generated from the labels alone: optionally a zoom on a jittered box around the lesion (always for lesions under 2% of the image), windowing on 30% of grey-scale images, then the expected final action with short template reasoning. Half of the templated questions are reworded with one of four training paraphrases; three further paraphrases are reserved for testing.

Stage 1, supervised fine-tuning. 16,000 views drawn once at random from the training pools, one pass (3,706 optimiser steps), learning rate 2×10^{-4}; cases with lesions under 2% of the image are repeated twice. **Stage 2, rejection-sampling fine-tuning** [27, 28]. The stage-1 model samples $k = 4$ episodes at temperature 1.2 on 200 training views per view kind, 500 lesion-free and 300 lesion-erased views; the best successful episode of each view is kept, and episodes from views the model mostly failed (at most two of four right) are repeated three times. Training continues on these plus 3,000 synthetic views at learning rate 5×10^{-5}. **Stage 3, general-VQA replay.** 5,000 views mixing MedScope views with ordinary questions from the *training* splits of VQA-RAD, SLAKE and PathVQA (any image near-identical to an evaluation image excluded), with "absent" views at no less than 35% of the mix and "Is X healthy?" replay questions balanced between yes and no; learning rate 5×10^{-5}.

3.6 Pre-registered selection

The data plan, recipe, selection rule and reporting rule were committed to the public repository on 1 November 2026, before any v4 data was built or model trained. Let U be validation net utility and H the validation rate of claimed findings on healthy images. Stage 3 is selected if $U_3 \geq U_2 - 0.02$ and $H_3 \leq H_2 + 0.02$; otherwise stage 2 under the same rule against stage 1; otherwise stage 1. Amendment 1 (after the stage-1 validation result, before any later result) also evaluated the previous release, v3, on the enlarged validation set, to be kept if the chosen stage fell more than 0.02 below it. Amendment 2 (before any test result) added comparisons that cannot change the selection.

On 540 validation cases (1,473 views), U and H were 0.585 and 13.6% for stage 1, 0.610 and 12.7% for stage 2, and 0.613 and 7.7% for stage 3; v3 scored 0.593 and 8.2%. The rule selected stage 3, which is MedScope-4B. It was then evaluated once on every test set; no setting, threshold or prompt was tuned afterwards.

3.7 Metrics

With O, T, R, N, D the original, target-removal, random-removal, natural-negative and degraded views:

$$\text{ESG} = P(\text{absent answer or refer} \mid T) - P(\text{absent answer or refer} \mid R).$$

CCR (covered-and-correct rate) is the share of answerable views ($O \cup R$) answered correctly; the **strict ESG** counts explicit absent answers only; **erased-lesion claims** is $P(\text{claims finding} \mid T)$ and **healthy-image claims** $P(\text{claims finding} \mid N)$; **grounded accuracy** is correct with IoU ≥ 0.3 on O; the **clarify rate** is $P(\texttt{clarify} \mid D)$; **net utility** is the mean reward over all views; **AURC** [29] uses the geometric mean of greedy answer-token probabilities as confidence.

3.8 Statistics

95% confidence intervals use 2,000 bootstrap resamples of cases [33]. Model comparisons use a paired bootstrap over shared cases with two-sided p-values. Seed results are reported as mean $\pm$ standard deviation over three seeds, with each seed pair also compared by paired bootstrap. Proportions on small subsets use Fisher's exact test. We do not adjust for multiple comparisons; the primary comparisons (ESG, erased-lesion claims and net utility against each frontier model on the main test) were fixed in advance, and all reach $p < 0.001$.

3.9 Evaluation harness

Open models decode greedily in bf16 on one A100 40 GB; API models use provider defaults. No model receives a model-specific repetition penalty. API models receive PNG images with the long side at most 768 pixels. Answers are parsed from the first final action; matching accepts paraphrases and health-question polarity and reads the asked finding first ("no pneumothorax" is no; "pneumothorax without mediastinal

Table 2: Main (internal) test set by source.

Source	Imaging	Cases with mask	Lesion-free	Split level
SLAKE [7]	CT / MRI	17	90	near-copy groups
Kvasir-SEG [34] + HyperKvasir [35]	Colonoscopy	83	91	near-copy groups
ChestX-Det [36]	Chest X-ray	41	50	patient
BUSI [38]	Breast ultrasound	42	15	near-copy groups
TCGA-LGG [39]	Brain MRI (FLAIR)	36	42	patient
Total		**219**	**288**	507 cases, 1,115 views

Table 3: Further test sets. "Fresh" sets were untouched until the model was selected (confirmatory).

Test set	Cases	Sources (cases)	Purpose
External centres	169	CVC-ClinicDB [40] (40), QaTa-COV19 [41] (20), BUS-BRA [42] (45), BraTS-Africa 2023 [43, 44] (64, with 50 tumour-free slices)	New scanners, populations
Fresh internal	76	ChestX-Det, unused test-split patients (36 with a lesion, 40 lesion-free)	Confirmatory
Fresh external	316	BraTS 2023 meningioma [45] (82), BUS-UCLM [46] (99), SIIM-ACR pneumothorax [47] (96), ETIS-LaribPolypDB [48] (39)	Confirmatory, new sources
Held-out wordings	373	Main test with three test-only paraphrases	Question wording
Standard VQA	1,051	VQA-RAD test (451), SLAKE English test (300 sampled), PathVQA test (300 sampled)	No regression

shift" is yes). A reply with no action tag is read as the model's decision, identically for every model, and that rate is reported. Correctness and rewards are recomputed from stored per-view transcripts by one script for every run.

4 Experimental setup

4.1 Study design and data

This is a retrospective study on public, de-identified datasets. The reference standard is each dataset's expert segmentation masks and image labels; expected answers are derived from them by rule (Section 3.2). No human annotation was used for training, selection or scoring.

Training and validation. Training uses SLAKE CT/MRI [7], Kvasir-SEG [34] with HyperKvasir normal landmarks as lesion-free images [35], ChestX-Det [36] (from NIH ChestX-ray14 [37]), BUSI [38] and TCGA-LGG [39]: 1,508 SLAKE and colonoscopy cases plus 1,792 chest X-ray, breast ultrasound and brain MRI cases after quality checks, with 690 lesion-free images from the latter three sources. Validation holds 540 cases (1,473 views). Diffuse chest findings, oversized lesions and brain slices with little tissue are excluded. Table 2 gives the main test set and Table 3 the further test sets, none of which was used for training or selection.

Protection of the test sets. When the v4 data were built, every existing test image was located in its source and checked to be identical (ChestX-Det 93, BUSI 57, TCGA-LGG 79 images; none differed or was missing), and every patient or near-copy group holding one was removed from all v4 splits. Patients seen in earlier training or validation were kept out of the fresh sets, and the fresh sets contain no near-copy of any training or test image. TCGA-LGG contributed no fresh internal cases because all 14 of its test-split patients already hold existing test images.

4.2 Models compared

Frontier models through their APIs, identical agent loop and provider defaults, evaluated on 31 October 2026: OpenAI gpt-6-sol, gpt-6-astra, gpt-6.1-sol; Anthropic Claude Fable 5.1, Claude Opus 5.5, Claude Sonnet 5.5; Google gemini-4-argon, gemini-3.1-pro-preview, gemini-3.8-flash. The three flagships (gpt-6-astra, Claude Opus 5.5, gemini-4-argon) were also run on the same 150 main-test cases with a cautious prompt, with four solved training examples in context, at their highest reasoning setting, and with held-out wordings. **Open models** (bf16): Qwen3.5-4B (MedScope's base) [31], MedGemma 1.5 4B [4], Lingshu-7B [5], HuatuoGPT-Vision-7B [2], Qwen2.5-VL-7B [49] and Qwen3-VL-8B [50], each with the agent prompt and, separately, with its own plain prompt. **Ablation arms**, all at stage 1 with the same 16,000-view draw, optimiser steps and learning-rate schedule: with counterfactual views (the recipe); without them (originals, degraded and lesion-free views only); without them but with lesion-free images repeated to match the recipe's share of "absent" answers; and erasures only (no edits elsewhere, no insertions). The first two are trained with three seeds each (seeds change LoRA initialisation, data order and trajectory sampling); both are repeated on MedGemma 1.5 4B. **Trivial policies** that ignore the image: always claim the finding, always say it is absent, always refer, always clarify.

4.3 Compute and software

Every training stage ran on a single NVIDIA A100 40 GB (Google Colab), about 4 hours per stage; one evaluation of a local model on the main test takes about 1 hour. Software: PyTorch 2.11, Transformers 5.22.1, PEFT 0.21.2, bitsandbytes 0.50.2. [Total API cost of all frontier runs, from the stored token counts.]

5 Results

5.1 Main test

MedScope-4B has the highest ESG (0.758) and the lowest erased-lesion claims (9.6%) of any model that answers (Table 4). It is better than all nine frontier and all six open models on ESG, erased-lesion claims and net utility (all $p < 0.001$; Table 5). Its accuracy on real lesions does not differ from any frontier model (CCR 0.849 vs 0.829–0.888; all $p \geq 0.12$). On unedited healthy images it reports a finding in 7.6% of cases, fewer than eight of nine frontier models ($p \leq 0.001$); against gemini-4-argon (9.0%) the difference is not significant ($p = 0.59$). It asks to clarify on every degraded view (170 of 170). Qwen3-VL-8B and Qwen2.5-VL-7B claim few erased lesions because they rarely answer lesion questions correctly at all (CCR 0.55 and 0.14); their ESG stays low.

5.2 By imaging type

The advantage holds in every imaging type (Table 6). MedScope-4B is strongest on colonoscopy (ESG 0.95), brain MRI and breast ultrasound (0.74–0.75), and weakest on chest X-ray (0.51) and SLAKE CT/MRI (0.47). No flagship exceeds ESG 0.69 in any type, and on chest X-ray the three flagships lie between -0.05 and 0.20.

5.3 What produces the behaviour

The counterfactual views are the active ingredient. Removing them from stage-1 training, with the same data draw, optimiser steps and schedule, lowers ESG on every seed (seed by seed $+0.525$, $+0.539$ and $+0.324$ in favour of the recipe, all $p < 0.001$) and raises erased-lesion claims to the frontier level or above (Table 7). The arm without counterfactual views says "present" more readily, so it is more accurate on real lesions (CCR 0.955 vs 0.871) and fails whenever the lesion is gone. Healthy-image claims also fall with the recipe (13.9% vs 35.9%), although both arms train on the same lesion-free images.

Table 4: Main test (507 cases, 1,115 views; agent prompt for every model). Erased, Healthy: % of views on which the model claims the finding after the lesion was erased, or on an unedited lesion-free image.

Model	CCR	ESG	Erased %	Healthy %	Net utility
MedScope-4B (this work)	0.849	**0.758**	**9.6**	**7.6**	**0.732** [CI]
gemini-4-argon	0.874	0.470	41.1	9.0	0.565
Claude Sonnet 5.5	0.870	0.402	48.4	16.7	0.381
gemini-3.1-pro-preview	0.849	0.384	51.6	26.7	0.306
gpt-6-sol	0.829	0.342	51.6	25.7	0.261
Claude Fable 5.1	0.856	0.342	52.1	23.3	0.255
Claude Opus 5.5	0.863	0.329	53.9	23.6	0.358
gpt-6.1-sol	0.888	0.324	58.9	30.9	0.276
gpt-6-astra	0.886	0.306	58.9	27.8	0.309
gemini-3.8-flash	0.868	0.269	61.6	27.8	0.263
Lingshu-7B	0.527	0.416	32.0	12.5	−0.249
Qwen3.5-4B (base)	0.758	0.338	42.0	19.4	0.134
Qwen3-VL-8B	0.553	0.297	12.8	12.5	0.186
MedGemma 1.5 4B	0.674	0.283	54.3	24.0	−0.255
HuatuoGPT-Vision-7B	0.498	0.265	30.1	22.9	−0.426
Qwen2.5-VL-7B	0.144	0.146	5.5	24.7	−0.742
Always claims the finding	1.000	0.000	100	100	−0.478
Always says it is absent	0.000	0.000	0	0	−0.848
Always refers	0.000	0.000	0	0	0.110

Table 5: MedScope-4B minus each model, paired bootstrap over the same 507 cases: difference [95% CI]. Positive is better for ESG and net utility; negative is better for the claim rates. Against the other four open models every ESG and net-utility difference is also $p < 0.001$.

Compared with	ESG	Erased-lesion claims	Healthy-image claims	CCR	Net utility
gemini-4-argon	+0.288 [0.208, 0.367]	−0.315 [−0.385, −0.243]	−0.014 [−0.059, 0.028]	−0.025 [−0.070, 0.017]	+0.168 [0.083, 0.254]
Claude Sonnet 5.5	+0.356 [0.284, 0.431]	−0.388 [−0.459, −0.318]	−0.090 [−0.137, −0.045]	−0.021 [−0.069, 0.026]	+0.351 [0.267, 0.437]
gemini-3.1-pro-prev.	+0.374 [0.298, 0.448]	−0.420 [−0.491, −0.350]	−0.191 [−0.252, −0.133]	0.000 [−0.043, 0.044]	+0.426 [0.337, 0.519]
gpt-6-sol	+0.416 [0.341, 0.489]	−0.420 [−0.491, −0.346]	−0.181 [−0.235, −0.126]	+0.021 [−0.022, 0.063]	+0.472 [0.389, 0.554]
Claude Fable 5.1	+0.416 [0.339, 0.496]	−0.425 [−0.500, −0.355]	−0.156 [−0.212, −0.102]	−0.007 [−0.054, 0.039]	+0.477 [0.384, 0.566]
Claude Opus 5.5	+0.429 [0.345, 0.507]	−0.443 [−0.515, −0.368]	−0.160 [−0.217, −0.103]	−0.014 [−0.066, 0.036]	+0.374 [0.283, 0.466]
gpt-6.1-sol	+0.434 [0.354, 0.520]	−0.493 [−0.566, −0.423]	−0.233 [−0.292, −0.178]	−0.039 [−0.088, 0.010]	+0.456 [0.360, 0.552]
gpt-6-astra	+0.452 [0.364, 0.539]	−0.493 [−0.562, −0.426]	−0.201 [−0.261, −0.145]	−0.037 [−0.087, 0.014]	+0.424 [0.329, 0.522]
gemini-3.8-flash	+0.489 [0.407, 0.566]	−0.521 [−0.596, −0.446]	−0.201 [−0.259, −0.143]	−0.018 [−0.065, 0.029]	+0.470 [0.380, 0.560]
Qwen3.5-4B (base)	+0.420 [0.338, 0.498]	−0.324 [−0.398, −0.249]	−0.118 [−0.171, −0.070]	+0.091 [0.035, 0.148]	+0.599 [0.503, 0.695]
MedGemma 1.5 4B	+0.475 [0.391, 0.556]	−0.447 [−0.518, −0.376]	−0.163 [−0.223, −0.104]	+0.176 [0.116, 0.235]	+0.988 [0.878, 1.099]

Table 6: Main test by imaging type: ESG / erased-lesion claims (fraction) / net utility.

Imaging (cases)	MedScope-4B	gemini-4-argon	Claude Opus 5.5	gpt-6-astra
Colonoscopy (174)	**0.95** / 0.01 / 1.06	0.69 / 0.31 / 0.83	0.57 / 0.40 / 0.53	0.40 / 0.58 / 0.40
Brain MRI (78)	**0.75** / 0.14 / 0.75	0.42 / 0.47 / 0.56	0.31 / 0.61 / 0.51	0.50 / 0.31 / 0.43
Breast ultrasound (57)	**0.74** / 0.14 / 0.78	0.40 / 0.48 / 0.53	0.29 / 0.62 / 0.40	0.12 / 0.83 / 0.28
Chest X-ray (91)	**0.51** / 0.17 / 0.40	0.20 / 0.49 / 0.21	−0.05 / 0.66 / −0.12	0.15 / 0.66 / 0.14
SLAKE CT/MRI (107)	**0.47** / 0.12 / 0.22	0.35 / 0.41 / 0.38	0.24 / 0.59 / 0.32	0.29 / 0.47 / 0.17

It is not the share of "No" answers. Repeating lesion-free images until the share of "absent" targets equals the recipe's leaves ESG at 0.215 (recipe 0.735; difference +0.521 [0.440, 0.600], $p < 0.001$).

The control edit matters. Training on erasures alone (no edit elsewhere, no insertions) reaches a similar ESG on the main test (0.708 vs 0.735, $p = 0.37$) by denying more: it claims fewer erased lesions (6.8% vs 11.9%, $p = 0.019$) but is less accurate on real lesions (CCR 0.788 vs 0.847, $p = 0.002$). The cost shows on an unseen tool: after a Telea edit elsewhere it denies the intact lesion in 43% of cases (recipe 23%, $p < 0.001$), so without the control a model partly learns to react to edits (Section 5.4).

It replicates on a second base model. The same recipe on MedGemma 1.5 4B gives ESG 0.721 vs

Table 7: Stage-1 arms on the main test (507 cases). Three-seed arms: mean ± sd.

Arm	CCR	ESG	Erased %	Healthy %	Net utility
Qwen3.5-4B, with counterfactual views (3 seeds)	0.871 ± 0.022	$\mathbf{0.741 \pm 0.019}$	13.2 ± 3.2	13.9 ± 4.0	$\mathbf{0.684 \pm 0.021}$
Qwen3.5-4B, without (3 seeds)	0.955 ± 0.008	0.279 ± 0.107	67.6 ± 11.9	35.9 ± 3.9	0.341 ± 0.090
Without, "absent" share matched (seed 0)	0.945	0.215	73.5	31.6	0.321
Erasures only (seed 0)	0.788	0.708	6.8	5.9	0.667
MedGemma 1.5 4B, with counterfactual views	0.806	0.721	7.8	5.9	0.698
MedGemma 1.5 4B, without	0.947	0.374	57.5	31.6	0.409
MedGemma 1.5 4B, untrained	0.674	0.283	54.3	24.0	−0.255

Table 8: Edited views by inpainting tool, 125 cases: ESG / erased-lesion claims (%).

Model	LaMa (training)	Transplant (training)	Telea (unseen)	Stable Diffusion (unseen)
MedScope-4B	**0.752** / 7.2	**0.720** / 8.0	**0.664** / 7.2	0.640 / 17.6
gemini-4-argon	0.536 / 32.8	0.488 / 32.8	0.560 / 8.8	0.392 / 40.0
Claude Opus 5.5	0.376 / 45.6	0.264 / 56.8	0.296 / 41.6	0.352 / 46.4
gpt-6-astra	0.312 / 55.2	0.240 / 64.0	0.256 / 60.8	0.256 / 61.6
Stage 1, with counterfactual views	0.720 / 12.0	0.744 / 8.0	0.696 / 7.2	0.568 / 24.8
Stage 1, erasures only	0.688 / 7.2	0.656 / 4.8	0.544 / 2.4	**0.712** / 4.8
Stage 1, without counterfactual views	0.272 / 69.6	0.200 / 76.8	0.144 / 83.2	0.232 / 73.6
Stage 1, without, "absent" share matched	0.224 / 72.0	0.208 / 73.6	0.096 / 85.6	0.184 / 77.6

0.374 without counterfactual views (+0.347 [0.276, 0.425], $p < 0.001$) and erased-lesion claims 7.8% vs 57.5%. With counterfactual views the two base models do not differ (net utility −0.005, $p = 0.83$; ESG +0.014, $p = 0.73$).

Stages 2 and 3 add little on the main test. The final model against its own stage 1: net utility +0.040 ($p = 0.15$), ESG +0.023 ($p = 0.39$), healthy-image claims −1.7 points ($p = 0.35$). Their contribution is on validation healthy-image claims, which the selection rule used (13.6% → 7.7%), and on ordinary VQA questions (Section 5.6).

The recipe's effect also holds at the external centres (three seeds: ESG 0.622 ± 0.030 vs 0.389 ± 0.043; erased-lesion claims 14.6% vs 48.2%; healthy-image claims 9.3% vs 49.3%) and on the fresh external test (net utility 0.070 ± 0.008 vs -0.149 ± 0.033; healthy-image claims 9.7% vs 26.1%), where the ESG difference is smaller (0.381 vs 0.288) and significant on two of three seeds.

5.4 Does it read the lesion or the edit?

Every training erasure used LaMa or tissue transplant. If MedScope-4B had learned those tools' traces, its advantage should vanish on other tools. We remade both edited views of 125 main-test cases on the same masks with Telea [18] and Stable Diffusion inpainting [20], neither used in training or selection, and with tissue transplant (Table 8).

MedScope-4B beats all three flagships on every tool (ESG differences +0.10 to +0.48, $p \leq 0.049$). On Telea its erased-lesion claims are unchanged (7.2%; ESG −0.088, $p = 0.025$, from more denials after the edit elsewhere); on Stable Diffusion they rise by 10.4 points ($p = 0.006$), the largest drop, yet still two to four times below Claude Opus 5.5 and gpt-6-astra. The arms without counterfactual views stay near ESG 0.1–0.3 on every tool. gemini-4-argon's erased-lesion claims fall from 33% to 9% on Telea only, whose smooth fills it recognises as edits: its absent answers partly follow visible traces.

Clean-looking erasures. Asked for the probability that an image was edited, gemini-4-argon's ratings of the 219 erased views are bimodal (61 rated ≤ 0.20, 158 rated ≥ 0.75, none in between), so the split needs no threshold choice. On erasures it rates as unedited, MedScope-4B still claims the lesion in only 15% (8% on visible ones), against 41–64% for the nine frontier models, which claim the lesion as often on visible erasures (41–58%).

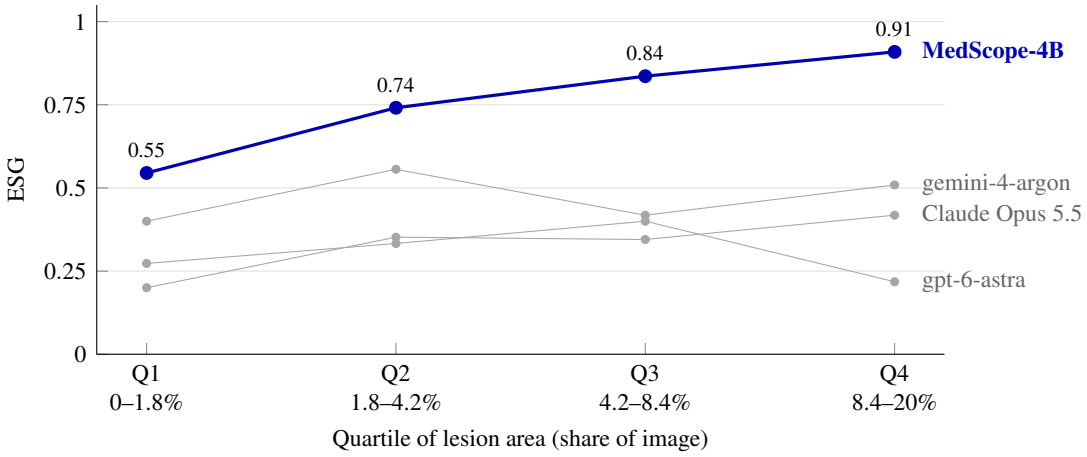


Figure 3: Evidence sensitivity gap by quartile of lesion area on the main test (55, 54, 55, 55 cases with a lesion mask). The gap between MedScope-4B and the flagships grows with lesion size but holds in the smallest quartile (0.55 vs 0.20–0.40).

Table 9: Flagships under four prompting conditions, same 150 cases: ESG / erased-lesion claims (%). MedScope-4B on these cases: 0.750 / 7.4.

Model	Default	Cautious prompt	Four solved examples	Highest reasoning
gemini-4-argon	0.441 / 41.2	0.412 / 42.6	0.456 / 36.8	0.456 / 39.7
Claude Opus 5.5	0.338 / 54.4	0.382 / 44.1	0.309 / 52.9	0.397 / 51.5
gpt-6-astra	0.338 / 52.9	0.397 / 47.1	0.500 / 38.2	0.324 / 58.8

Edit visibility. A ResNet-18 trained on 5,126 training views separates random-removal from original views with AUC 0.83 [0.80, 0.87], so the inpainting leaves a trace that a trained detector can find. Zero-shot, gemini-4-argon and gpt-6.1-sol cannot separate random-removal views from originals (AUC 0.52–0.55) but find erased-lesion views suspicious (0.70–0.75): erasing a lesion leaves a smooth region where structure is expected.

Four results argue that MedScope's answers follow the lesion rather than edit traces: (i) an edit detector scores ESG 0 by construction, and MedScope-4B denies the lesion on only 14.6% of random-removal views, which carry the same traces; (ii) the vision encoder is frozen, so the adapter can only reweight features the pretrained encoder already produces and cannot learn a pixel-level detector of the kind the ResNet-18 learns; (iii) the advantage holds on clean-looking erasures and on two unseen tools; (iv) on completely unedited lesion-free images, where no edit hypothesis applies, the recipe lowers false findings from 35.9% to 13.9% (Table 7).

5.5 Robustness

Lesion size. Evidence sensitivity rises with lesion area, and the smallest lesions remain the weakest (Figure 3; erased-lesion claims 18.2% in the smallest quartile and 1.8% in the largest). The flagships show no such trend: their ESG stays between 0.20 and 0.56 in every quartile, and gpt-6-astra claims a large erased lesion in 75% of cases.

Prompting the frontier models harder. MedScope learned from thousands of counterfactual views; the frontier models saw none. On the same 150 cases we gave the three flagships a cautious prompt (an explicit warning that the question may name an absent finding), four solved training examples in context (a lesion; the same image erased; an edit elsewhere with the lesion intact; a healthy image), and their highest reasoning setting. None closes the gap (Table 9). At the highest reasoning setting no flagship differs from its default (all $p \geq 0.13$). MedScope-4B stays better than every flagship under every condition on ESG ($p \leq 0.001$) and net utility ($p \leq 0.034$).

Table 10: MedScope-4B and the three flagships on the further test sets.

Test set	Model	CCR	ESG	Erased %	Healthy %	Net utility
External centres (169)	**MedScope-4B**	0.727	**0.597**	**11.8**	**4.0**	0.530
	gemini-4-argon	0.950	0.370	58.8	18.0	0.544
	Claude Opus 5.5	0.966	0.412	53.8	30.0	0.544
	gpt-6-astra	0.945	0.277	66.4	24.0	0.424
Fresh internal (76)	**MedScope-4B**	0.681	**0.472**	**19.4**	**12.5**	**0.402**
	gemini-4-argon	0.639	0.056	61.1	12.5	0.048
	Claude Opus 5.5	0.653	0.000	63.9	25.0	−0.023
	gpt-6-astra	0.847	0.083	75.0	17.5	0.219
Fresh external (316)	**MedScope-4B**	0.476	0.393	**8.3**	**4.7**	−0.053
	gemini-4-argon	0.741	0.435	31.5	9.5	**0.310**
	Claude Opus 5.5	0.696	0.292	42.3	18.9	0.056
	gpt-6-astra	0.634	0.268	35.7	34.5	−0.106

Open models with their own prompt. With a short native prompt and one plain reply, every open model claims erased lesions more often than under the agent prompt (53–72%) [regenerate with the v4-era runs], so the agent loop does not handicap them on this task.

Question wording. On 373 cases with three test-only paraphrases, net utility is unchanged against the original wording (+0.006 [−0.033, 0.041], $p = 0.74$). With the same paraphrases on 150 cases, MedScope-4B leads the flagships by +0.35 to +0.57 in net utility ($p < 0.001$).

Tools off. With every tool call refused at test time, ESG is unchanged (0.753 vs 0.758, $p = 0.92$) and net utility falls by 0.092 ($p = 0.003$): the model says "absent" more often on lesion views (21% vs 15%). Evidence sensitivity comes from training; the tools add accuracy on real lesions. MedScope-4B calls a tool on 0.99 of lesion views (35% of calls are windowing), and its first zoom covers the lesion in 93%. gpt-6-sol and gpt-6-astra almost never call a tool.

Cost matrix. The always-refer policy earns +0.110 under the published matrix, a weakness of that matrix; with referral on lesion-free images paid 0 it earns −0.118, and −0.300 when referral costs −0.3 everywhere. MedScope-4B ranks first among all models under all five matrices (wrong-answer penalty −2, −1, −0.5; the two referral variants), and its strict ESG equals its ESG (0.758), since it never refers.

Degraded views. Without the degraded views, which favour a model trained to clarify, net utility is 0.774 for MedScope-4B and 0.310–0.579 for the flagships (MedScope better, $p < 0.001$).

Label noise. Without human review, the nine frontier models served as an independent panel: a lesion-free test image that at least seven call abnormal is treated as a disputed label (26 of 288: 12 colonoscopy, 11 SLAKE, 2 breast ultrasound, 1 chest X-ray; MedScope does not vote). Without them, MedScope-4B's healthy-image claims are 6.9% (gemini-4-argon 4.6%, other frontier models 10.3–24.4%), and every comparison in Table 5 keeps its direction and significance.

5.6 Generalisation

On every new test set MedScope-4B keeps the lowest erased-lesion claims and the lowest (once tied) healthy-image claims of the four models, but the flagships detect more real lesions (Table 10). Net utility, which weighs both, does not differ at the external centres, favours MedScope-4B on the fresh internal set, and favours gemini-4-argon on the fresh external set.

External centres. ESG is higher than every flagship's ($p \leq 0.002$), erased-lesion claims are 42–55 points lower ($p < 0.001$) and healthy-image claims 14–26 points lower ($p \leq 0.026$), while CCR is 0.22–0.24 lower ($p < 0.001$); net utility does not differ ($p \geq 0.07$). By source, ESG is 0.85 on CVC-ClinicDB, 0.71 on BraTS-Africa, 0.58 on BUS-BRA and 0.05 on QaTa-COV19, where diffuse COVID-19 opacities are erased far less completely than focal lesions.

Fresh internal (confirmatory, chest X-ray). ESG 0.47 against 0.00–0.08 for the flagships ($p \leq 0.001$);

Table 11: Fresh external test by source: CCR / net utility.

Source (cases)	MedScope-4B	gemini-4-argon	Claude Opus 5.5	gpt-6-astra
BUS-UCLM, breast ultrasound (99)	0.73 / **0.67**	0.71 / 0.55	0.93 / 0.48	0.98 / 0.18
ETIS-Larib, colonoscopy (39)	0.69 / 0.38	0.97 / **0.75**	0.92 / 0.59	0.73 / 0.35
BraTS 2023 meningioma, MRI (82)	0.53 / −0.03	0.62 / **0.09**	0.69 / 0.00	0.50 / −0.21
SIIM-ACR pneumothorax, chest X-ray (96)	0.00 / −1.10	0.66 / **−0.06**	0.28 / −0.68	0.29 / −0.62

Table 12: Standard medical VQA, same agent loop for every model: closed / open / all questions.

Model	VQA-RAD (451)	SLAKE (300)	PathVQA (300)
MedScope-4B	0.741 / 0.230 / 0.514	0.881 / 0.644 / 0.730	0.801 / 0.125 / 0.477
gemini-4-argon	0.884 / 0.410 / 0.674	0.927 / 0.696 / 0.780	0.712 / 0.097 / 0.417
Claude Opus 5.5	0.876 / 0.380 / 0.656	0.927 / 0.602 / 0.720	0.865 / 0.153 / 0.523
gpt-6-astra	0.797 / 0.255 / 0.557	0.835 / 0.476 / 0.607	0.526 / 0.062 / 0.303
Qwen3.5-4B (base)	0.657 / 0.260 / 0.481	0.826 / 0.424 / 0.570	0.545 / 0.021 / 0.293
Lingshu-7B	0.705 / 0.195 / 0.479	0.798 / 0.545 / 0.637	0.712 / 0.042 / 0.390

Table 13: SIIM pneumothorax images called present (the 42 cases found in the original-resolution data).

Model	256-pixel copy	Original resolution
Qwen3.5-4B (base)	0%	12%
Stage 1 without counterfactual views	0%	5%
Stage 1 with counterfactual views	19%	40%
Stage 2	[–]	40%
MedScope-4B (stage 3)	**0%**	**0%**
gemini-4-argon	69%	69%

net utility better than Claude Opus 5.5 and gemini-4-argon ($p = 0.002$), not different from gpt-6-astra ($p = 0.15$).

Fresh external (confirmatory). The result is mixed. ESG is higher than gpt-6-astra's and Claude Opus 5.5's ($p \leq 0.037$) and not different from gemini-4-argon's ($p = 0.34$); net utility is below gemini-4-argon's (-0.363 [-0.496, -0.233], $p < 0.001$). The by-source breakdown (Table 11) locates the loss: MedScope-4B leads on breast ultrasound and trails on colonoscopy and meningioma MRI, and on SIIM pneumothorax it answers "No" to every one of 48 pneumothorax images (Section 6).

Standard medical VQA. Accuracy on ordinary questions is kept (Table 12). MedScope-4B saw these benchmarks' training splits during replay, so this shows the absence of a regression, not a like-for-like comparison; against its own base model it gains on all three. With their own plain prompt most open models score higher than in the agent loop (Lingshu-7B: 0.552, 0.803, 0.573), so the loop is a harder format for open models on ordinary VQA.

6 Failure analysis: why is pneumothorax missed?

MedScope-4B answered a confident "No" on all 48 SIIM-ACR pneumothorax images of the fresh external test (answer-token probability about 1.0). Scoring and parsing were checked and are correct. The analysis below was made after the test results, is labelled post hoc in the pre-registration, and changes no model or confirmatory number. It trains nothing: every number comes from stored runs or from rebuilding each stage's exact training list with the training code.

Resolution is not the cause. The fresh test used a 256-pixel copy of SIIM. Rebuilt from the original DICOMs, earlier checkpoints detect more pneumothoraxes, but the final model still detects none (Table 13).

Counterfactual training is not the cause. The stage-1 model with counterfactual views is the most sensitive Qwen-based model at both resolutions; the arm without them detects almost none.

Training coverage is. Questions naming pneumothorax were 54 of about 7,000 chest X-ray training views in stage 1 (35 in stage 2, 15 in stage 3), trained to "No" 80%, 100% and 87% of the time. Stage 1 held only four pneumothorax lesion images, all large (lesion box median 37% of the image, against 6.9% for the SIIM cases). No validation or internal test set held a pneumothorax lesion image asked about pneumothorax (only 7, 7 and 4 lesion-free ones), so the selection rule could not see the problem.

Stage 3 turns weak detection into none. Stage 3 shifts chest X-ray answers towards "No" in general: on validation, named findings detected fall from 63% to 56%, erased-lesion claims from 16% to 9% and healthy-image claims from 24% to 10%. With almost no positive pneumothorax examples, its weak detection drops to zero at every lesion size. The share of "No" targets alone does not explain this: stage 2 trained on 35 of 35 "No" targets and still detected 40%.

The model registers something. Asked "Is there an abnormality?", MedScope-4B says yes on 35 of 42 pneumothorax images and on 28 of 48 SIIM images without pneumothorax (Fisher $p = 0.012$); the base model separates them more (38/42 vs 23/48, $p < 0.001$). The final model sees that something is abnormal but does not name pneumothorax.

Lessons. A rare critical finding needs positive examples in training and in validation, and the selection rule needs a per-finding sensitivity floor; otherwise a conservative final stage can remove it entirely while every aggregate metric improves. Coverage may not be the whole story: the gap to gemini-4-argon even at full resolution is consistent with the frozen vision encoder limiting subtle findings.

7 Reader study

The benchmark's labels come from datasets, and the erasures are checked only automatically. Physicians therefore read a blinded set of test images to check two assumptions: that an erased finding is no longer visible, and that erased images are not trivially recognisable as edited. [Results pending; this section gives the design and the analysis fixed before any answer was seen.]

Images. 96 images from the main test, each from a different case and stratified by source: 32 unmodified images with a lesion, 32 lesion-erased images (the LaMa erasures the models were tested on) and 32 unmodified lesion-free images. Lesion-free images are included so that "no lesion" is not a cue for "edited". Only yes/no presence questions are used. Every image is re-saved the same way (PNG, no metadata) under an opaque name.

Procedure. Readers use a static web page that holds no answer key, in their own random order, with no time limit and no other information. For each image they answer (1) whether the image has been digitally edited (definitely real, probably real, probably edited, definitely edited) and (2) the case's clinical question (yes, no, cannot tell). They see no results, no paper and no image counts by kind until their answers are submitted.

Pre-specified analysis. Per reader: the AUC for separating erased from unmodified images on the four-level rating, and the share of erased images rated probably or definitely edited; the share answering "yes" to the clinical question on each image kind; agreement between readers (Cohen's κ on edited yes/no); and results by source. The benchmark's assumptions are supported if readers rarely see the erased finding, and if their AUC for recognising erasures is close to the zero-shot frontier models' (0.70–0.75) rather than near 1.

Readers. [Number, specialty, years of experience.] **Results.** [Pending.]

8 Discussion

8.1 Principal findings

Seeing a lesion and reporting only what is seen are separable abilities. Nine frontier models see real lesions as well as a 4B model, yet claim about half of the lesions after they have been erased, under the

benchmark prompt, under a cautious prompt, with solved examples in context, at their highest reasoning setting, and with reworded questions. A small model acquires the disposition to report only what is visible from automatically built counterfactual views, and the ablations identify those views — not the data, the answer format or the share of "No" answers — as the active ingredient. The effect holds on every seed, on a second base model, and on inpainting tools never used in training. Because the views come from public masks and automatic checks, the recipe extends to any dataset with segmentation labels.

8.2 What it is not

The gain is in evidence sensitivity, not in perception. At new centres the frontier models detect more real lesions (CCR 0.95–0.97 vs 0.73), and the two abilities pull net utility in opposite directions: frontier models behave as high-sensitivity, high-hallucination readers, MedScope-4B as an evidence-gated reader that misses more unfamiliar lesions but rarely reports absent ones. Which is preferable depends on the clinical cost of each error, which our cost matrix fixes only by assumption. The fresh external test shows the danger of the evidence-gated reader at its worst: where perception or training coverage is weak, conservatism becomes a complete miss of a critical finding.

8.3 Relation to prior work

Hallucination benchmarks and training-free detectors [9, 10, 11, 12, 13, 15, 16] measure or flag unsupported claims; we show that the same counterfactual idea, applied to the image with an edit-matched control, can train the behaviour into a model. Compared with natural-image counterfactual VQA [17], the control edit with the same tool and shape closes the shortcut of reacting to edits, and our erasures-only arm shows that without it a model partly learns that shortcut. Compared with tool-using agents trained by reinforcement learning [21, 22, 23, 24], label-derived trajectories and rejection sampling were enough to teach tool use (first zoom on the lesion in 93% of lesion views).

8.4 Implications

For developers, a model's answers to presupposing questions should be tested on images with the finding removed, not only on images where it is present. For evaluators, aggregate accuracy and even aggregate hallucination rates can hide a complete miss of one critical finding; per-finding sensitivity floors belong in model selection. For clinical use, MedScope-4B is a research model, not a medical device, and nothing here supports unsupervised use.

8.5 Limitations

- **Rare critical findings.** A finding almost absent from training and absent from validation was lost entirely by the most conservative stage (Section 6).
- **Perception.** The vision encoder is frozen; subtle findings and the smallest lesions (ESG 0.55 in the lowest quartile) remain weakest.
- **Erasure as ground truth.** Diffuse findings are erased less completely (QaTa-COV19 ESG 0.05), erasure can leave secondary signs such as mass effect or oedema, and a second, unmasked lesion (Figure 2, breast ultrasound) makes the expected "absent" answer wrong for every model. The reader study tests how often an erased finding remains visible.
- **Edit traces.** A trained detector finds the edits (AUC 0.83). The advantage holds on clean-looking erasures and unseen tools, but a residual contribution of edit traces cannot be excluded.
- **Reference standard.** Labels and masks come from public datasets without re-reading; disputed lesion-free labels (26 of 288) do not change any comparison.
- **Same construction for training and test.** MedScope trained on views built like the test views; the frontier models saw none. The ablation trained on the same data without counterfactual views hallucinates

at the frontier level, which isolates the views as the cause, and solved examples in context did not close the gap.
- **Frontier models** were evaluated through public APIs on fixed dates with default settings; their behaviour may change.
- **Scope.** Yes/no and short-answer questions on single 2D images; no report generation, no 3D volumes, no prospective or clinical evaluation, and limited demographic information in the source datasets.
- **No human annotation.** All checks are automated or model-based until the reader study is complete.

8.6 Future work

Three directions follow directly: per-finding coverage and sensitivity floors in data and selection; unfreezing or replacing the vision encoder so that evidence sensitivity is combined with frontier-level perception; and extending the views to report generation, where a removed finding should disappear from the report while the rest stays.

9 Conclusion

Erasing the lesion, with an edit of the same shape elsewhere as the control, turns "does the answer follow the image?" into a property that can be measured and trained without human annotation. Trained on such views and selected by a pre-registered rule, a 4B agent matches nine frontier models on real lesions and claims erased findings four to six times less often, robustly to prompting, question wording and unseen inpainting tools, while keeping ordinary VQA ability. It is only as good as the findings its training data covers: a pre-registered fresh test exposed a complete miss of pneumothorax, and the fix belongs in data coverage and model selection, not in prompting.

Declarations

Data and code availability. Code, view-building recipes, notebooks, the pre-registration with its git history, and the results log: `https://github.com/bhaskargurram-ai/medscope`. LoRA adapters and MedScope-Bench (view manifests and per-view transcripts of every evaluation run) on Hugging Face (`bhaskar-ai/MedScope-4B`; `datasets/bhaskar-ai/MedScope-Bench`), public on acceptance. All images come from public de-identified datasets under their own licences and are not redistributed.

Funding. This research received no external funding. **Competing interests.** The author declares no competing interests. **Ethics.** The model study used only public, de-identified datasets under their published licences; no new patient data were collected. [Reader study: participant consent and local ethics position.] **Author contributions (CRediT).** Bhaskar Gurram: conceptualisation, methodology, software, investigation, formal analysis, data curation, writing (original draft, review and editing), visualisation. [Reader-study physicians: acknowledged or credited as agreed.] **Declaration of generative AI use.** [To be completed according to the target journal's policy.]

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A Prompts

Agent prompt (every model, all main results):

```
You are MedScope, a careful medical imaging assistant used for research. You look at a medical
    image and answer a question about it from what the image actually shows: you report a finding
     only when you can see it, and you say so when it is not there.
On each turn, think briefly in plain text, then write exactly ONE action as a tagged JSON object
    and stop. After a tool call, stop and wait: you will be shown the result before you continue.
[two examples: a zoom turn and a final answer]
Tools (at most 3 calls per case; boxes are integers 0-1000 on the ORIGINAL full image, as [x1, y1
    , x2, y2]):
<tool>{"name": "zoom", "box": [x1, y1, x2, y2]}</tool> look closer at a region
<tool>{"name": "window", "center": 128, "width": 100}</tool> adjust brightness/contrast (0-255
    scale)
Final actions (exactly one, then stop):
```

```
<final>{"action": "answer", "answer": "...", "box": [x1, y1, x2, y2], "confidence": 0.0-1.0}</
    final> when the image lets you answer. If the finding is visible, the box marks it on the
    original image. If the finding asked about is not there, say so and leave out the box: "No"
    to "Is there ...?", "Yes" to "Is ... healthy?" or "Is ... normal?", "No abnormality" to an
    open question.
<final>{"action": "clarify", "question": "..."}</final> when the image is unusable (too blurred,
    noisy, low resolution, or not the anatomy asked about).
<final>{"action": "refer", "reason": "..."}</final> when the image is usable but you cannot
    decide either way.
Keep answers short, using the wording a radiologist would use. Never report a finding you cannot
    see.
```

Plain prompt (open models, own-prompt runs): "You are a medical imaging assistant. Look at the image and answer the question. Keep the answer short." One reply, no tools. **Cautious prompt** (Table 9): the agent prompt followed by "Important: a question may name a finding that is not in the image. Do not assume the question is correct. Before answering, check carefully whether the finding is actually visible. If you cannot clearly see it, say it is not there ("No"). Reporting a finding that is not visible is a serious error." **Few-shot condition** (Table 9): the cautious prompt plus four solved training examples before each case (a lesion; the same image with it erased; an edit elsewhere with the lesion intact; a healthy image). Example images never appear in any test set.

B Reproducibility

- **Hardware.** One NVIDIA A100 40 GB per training stage (about 4 hours each); one local model on the main test, about 1 hour. Frontier models run from a CPU runtime.
- **Determinism.** Greedy decoding for open models; the 16,000-view stage-1 draw is fixed across seeds; seeds change LoRA initialisation, data order and trajectory sampling.
- **Re-scoring.** Every table is regenerated from stored per-view transcripts by one script; the notebook behind each number is named in the results log.
- **Pre-registration.** `docs/PREREGISTRATION_v4.md`, committed before any v4 data existed; its git history records the rule, both amendments and both post-hoc notes.
- **Negative results from development.** Plain GRPO from a stage-1 agent gave reward differences in only 20% of sampled groups, so group-relative advantages vanished; dynamic sampling gained one validation case. Merging a grounding-only adapter into the agent removed evidence sensitivity (abstention gap 0.90 $\rightarrow$ 0.00) while keeping grounding. An adapter trained only to answer with boxes claimed the finding on every erased and every lesion-free image.

C CLAIM 2024 checklist (summary)

Item	Where addressed
Study design (retrospective); intended use (research)	Sections 1, 4.1; Discussion
Data sources, inclusion and exclusion	Section 4.1, Tables 2–3
De-identification	Public de-identified sources; no new patient data
Reference standard and its limits	Sections 3.2, 8.5; reader study (Section 7)
Partitions and disjointness	Patient or near-copy-group splits; test protection (Section 4.1)
Model, initialisation, training details	Section 3.5; Appendix B
Selection of the final model	Section 3.6 (pre-registered)
Metrics, uncertainty, statistical tests	Sections 3.7, 3.8
Robustness and sensitivity analyses	Sections 5.3–5.5
Internal and external testing	Sections 5.1, 5.6
Failure analysis	Section 6
Availability of code, model, data	Declarations